

Variable ratios — direct and inverse proportion

When one quantity grows with another, and when it shrinks instead.

LESSON 50–52

3 × 40 MIN

MATEMATIKA 6, §12

S7 12

LESSON CLOCK

25'

30'

30'

25'

10

Direct proportion	25 min
The unitary method	30 min
Inverse proportion	30 min
Which is which?	25 min
Homework	10 min

LEARNING OBJECTIVES

- Recognise direct proportion and use the unitary method.
- Recognise inverse proportion and use the constant product.
- Decide which kind a situation is before calculating.
- Read a proportional relationship from a table.

TEXTBOOK

National	pp. 141–150
Cambridge	Stage 7, pp. 118–125
Workbook	Exercise 12.5

01

Explanation

Direct proportion

Two quantities are in **direct proportion** when doubling one doubles the other. Their quotient stays the same.

$$\frac{y}{x} = k, \text{ a constant}$$

PENS	1	2	5	8
cost, sum	3000	6000	15 000	24 000
cost ÷ pens	3000	3000	3000	3000

THE CONSTANT IS THE PRICE OF ONE

In every direct proportion the constant has a meaning: the cost of one pen, the distance in one hour, the mass of one metre. Finding it is the whole method.

The unitary method

PROBLEM	ONE UNIT	ANSWER
5 pens cost 15 000: 8 pens	3000	24 000
4 m of cloth costs 60 000: 7 m	15 000	105 000
a car does 180 km in 3 h: in 5 h	60 km	300 km
6 identical books weigh 2.4 kg: 10 books	0.4 kg	4 kg

DIVIDE, THEN MULTIPLY

Down to one, then up to what is asked. Two operations, in that order, solve every direct-proportion problem in the course.

Inverse proportion

Two quantities are in **inverse proportion** when doubling one halves the other. Their product stays the same.

$xy = k, \text{ a constant}$

WORKERS	2	3	4	6
days	12	8	6	4
product	24	24	24	24

THE CONSTANT IS THE SIZE OF THE JOB

24 worker-days is the work itself, whoever does it. Once that number is found, any number of workers can be answered by dividing.

Which is which?

SITUATION	KIND	CONSTANT
cost of n identical pens	direct	the price of one
distance in t hours at a fixed speed	direct	the speed
workers and days for one job	inverse	worker-days
speed and time for a fixed distance	inverse	the distance
taps and time to fill a tank	inverse	tap-minutes
a person's age and height	neither	—

NOT EVERYTHING THAT CHANGES TOGETHER IS PROPORTIONAL

A child twice as old is not twice as tall, and eight workers may get in each other's way. Proportion is a model, and the question is always whether it fits.

02 Worked examples

EXAMPLE 1 5 pens cost 15 000 sum. What do 8 cost?

- 1 More pens, more money — direct.
- 2 One pen: $15\,000 \div 5 = 3000$.
- 3 Eight pens: $8 \cdot 3000 = 24\,000$.

ANSWER

24 000 sum

EXAMPLE 2 4 workers finish a job in 6 days. How long would 8 take?

① More workers, fewer days — inverse.

② $4 \cdot 6 = 24$ worker-days.

③ $24 \div 8 = 3$ days.

ANSWER

3 days

EXAMPLE 3 A car covers 180 km in 3 hours at a steady speed. How far in 5 hours?

① Direct: more time, more distance.

② One hour: 60 km.

③ Five hours: 300 km.

ANSWER

300 km

03 Interactive model

Ask how long six taps take if three take twenty minutes; the wrong answer of forty is worth hearing aloud before the constant product is introduced.

Direct, inverse, or neither?

The cost of n identical pens:

Workers and days for one job:

5 pens cost 15 000; 8 cost:

4 workers take 6 days; 8 take:

12

3

2

4

Speed and time for a fixed distance:

direct

inverse

neither

constant

In direct proportion the constant is:

xy

$\frac{y}{x}$

$x + y$

$x - y$

In inverse proportion the constant is:

$$xy$$

$$\frac{y}{x}$$

$$x + y$$

$$y - x$$

A person's age and height are:

directly proportional

inversely proportional

not proportional

constant

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Direct proportion	To'g'ri proporsionallik	Прямая пропорциональность
Inverse proportion	Teskari proporsionallik	Обратная пропорциональность
Unitary method	Birlik usuli	Метод приведения к единице
Constant	O'zgarmas	Постоянная
Product	Ko'paytma	Произведение
Rate	Tezlik, narx	Норма
To increase	Ortmoq	Увеличиваться
To decrease	Kamaymoq	Уменьшаться

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Direct proportion keeps constant:

the product

the quotient

the sum

the difference

Inverse proportion keeps constant:

the product

the quotient

the sum

the difference

4 m of cloth costs 60 000; 7 m costs:

90 000

105 000

120 000

84 000

6 taps fill a tank in 4 h; 8 taps take:

3 h

5 h

2 h

6 h

The unitary method means:

multiply then divide

divide to one, then multiply

add

take the product

Doubling one quantity halves the other: this is:

direct

inverse

neither

impossible

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 5 pens cost 15 000: one pen

ANSWER 3000

2. So 8 pens cost

ANSWER 24 000

3. 4 m of cloth costs 60 000: 7 m

ANSWER 105 000

4. 180 km in 3 h: in 5 h

ANSWER 300 km

5. 6 books weigh 2.4 kg: 10 books

ANSWER 4 kg

6. Cost against number of pens is

ANSWER direct

7. Workers against days is

ANSWER inverse

Medium · the standard question

7 PROBLEMS

1. 4 workers take 6 days: 8 workers

ANSWER 3 days

2. 6 taps fill a tank in 4 h: 8 taps

ANSWER 3 h

3. 3 taps take 20 min: 6 taps

ANSWER 10 min

4. Speed and time for a fixed distance

ANSWER inverse

5. Distance and time at a fixed speed

ANSWER direct

6. Age and height

ANSWER neither

7. 12 workers take 5 days: 4 workers

ANSWER 15 days

Hard · stretch and reason

7 PROBLEMS

1. If y is inversely proportional to x and $y = 12$ at $x = 2$: y at $x = 8$

ANSWER 3

2. If y is directly proportional to x and $y = 12$ at $x = 2$: y at $x = 8$

ANSWER 48

3. 9 workers take 8 days; how many for 6 days?

ANSWER 12

4. Food lasts 30 people 12 days: for 40 people

ANSWER 9 days

5. 15 m of wire weighs 600 g: the mass of 4 m

ANSWER 160 g

6. Why is “more workers, more days” never the model?

ANSWER The job is fixed, so the product must be

7. A tap fills a tank in 12 min: two such taps together

ANSWER 6 min

07 Homework

Homework — 5 tasks

Name the kind of proportion before doing any arithmetic.

- 7 notebooks cost 42 000 sum. Find the cost of 11.
- A car travels 240 km in 4 hours. How far in 7 hours?
- 5 workers finish a job in 12 days. How long would 10 take?
- 4 taps fill a tank in 15 minutes. How long would 6 take?

5. Say which of these is direct, which inverse and which neither: cost and quantity; speed and time; a pupil's age and marks.